

A cosecant certificate for the bidisc weak-product constant

Automated mathematics run `fa_banach_001`, lane 0

21 July 2026

Status: candidate partial result, likely valid. The result below improves a quantitative obstruction but does not resolve the equality of the two spaces.

1 Source question

Bampouras and Brevig ask whether, for fixed $d \geq 2$, there is a constant $C_d > 0$ such that

$$\|f\|_{H^1(\mathbb{T}^d)} \geq C_d \|f\|_{W(\mathbb{T}^d)}, \quad W(\mathbb{T}^d) = H^2(\mathbb{T}^d) \odot H^2(\mathbb{T}^d),$$

for every $f \in W(\mathbb{T}^d)$; equivalently, whether $H^1(\mathbb{T}^d)$ and $W(\mathbb{T}^d)$ are equal as sets [1]. The question occurs on page 3 of the source paper.

For $d \geq 2$, it is an important open problem in harmonic analysis (see [8]) whether there is an absolute constant $C_d > 0$ such that $\|f\|_{H^1(\mathbb{T}^d)} \geq C_d \|f\|_{W(\mathbb{T}^d)}$ for every f in $W(\mathbb{T}^d) = H^2(\mathbb{T}^d) \odot H^2(\mathbb{T}^d)$.

The work of Ortega-Cerdà and Seip discussed above shows that $C_2 \leq 2\sqrt{2}/\pi < 1$. A minor improvement can be found in [4 Theorem 5]. Since plainly

$$(3) \quad \|f\|_{H^1(\mathbb{T}^d)} \leq \|f\|_{W(\mathbb{T}^d)} \leq \|f\|_{H^2(\mathbb{T}^d)},$$

the open problem is to ascertain whether $H^1(\mathbb{T}^d)$ and $W(\mathbb{T}^d)$ are equal as sets. Note that (3) also shows that if φ is norm attaining in $H^1(\mathbb{T}^d)$, then φ is norm attaining in $W(\mathbb{T}^d)$. This inspires us to compare the admissible pairs $(H^2(\mathbb{T}^2), H^1(\mathbb{T}^2))$ and $(H^2(\mathbb{T}^2), W(\mathbb{T}^2))$ in detail. Our case study is concerned with the polynomials

$$\varphi_\alpha(z) = z_1^2 + \alpha z_1 z_2 + z_2^2$$

for $\alpha \geq 0$. In order to state our result, we let $\alpha_0 = 1.62420\dots$ denote the unique (see Lemma [6]) solution of the equation

$$\sqrt{4 - \alpha^2} = \frac{2}{\alpha} \arcsin \frac{\alpha}{2}$$

Figure 1: The open question in arXiv:2310.20346, page 3.

For the bidisc, put

$$K_2 := \sup_{0 \neq f \in W(\mathbb{T}^2)} \frac{\|f\|_{W(\mathbb{T}^2)}}{\|f\|_{H^1(\mathbb{T}^2)}} \in [1, \infty].$$

Thus the best possible constant in the displayed inequality is K_2^{-1} when $K_2 < \infty$. The quadratic construction used previously gives a lower bound about 1.16064 for K_2 [2]; the source paper also discusses the earlier Ortega–Cerdà–Seip obstruction [3].

2 Proof intuition

The source paper turns the weak-product dual norm into the operator norm of a small Hankel operator. For a homogeneous bidisc symbol, that Hankel operator is a direct sum of finite Hankel matrices. We choose the symbol coefficients from a discrete cosecant matrix which is exactly orthogonal. Every required Hankel block is then a submatrix of an isometry, giving a norm-one dual certificate at once.

Pairing this certificate with a homogeneous Dirichlet polynomial yields a quantity asymptotic to $(2/\pi) \log N$. The H^1 norm of that polynomial is the usual Dirichlet-kernel L^1 norm, asymptotic to $(4/\pi^2) \log N$. The quotient of the two leading constants is $\pi/2$. This supplies a quantitative partial result only; it neither proves nor disproves that K_2 is finite.

3 Statement

Theorem 1. *Let $W(\mathbb{T}^2) = H^2(\mathbb{T}^2) \odot H^2(\mathbb{T}^2)$. Then*

$$K_2 \geq \frac{\pi}{2}.$$

Consequently, if a constant $C_2 > 0$ satisfies $\|f\|_{H^1(\mathbb{T}^2)} \geq C_2 \|f\|_{W(\mathbb{T}^2)}$ for every $f \in W(\mathbb{T}^2)$, then

$$C_2 \leq \frac{2}{\pi}.$$

More precisely, for $N \geq 2$ define

$$f_N(z_1, z_2) = \sum_{k=0}^{N-1} z_1^k z_2^{N-1-k}$$

and

$$S_N = \frac{1}{N} \sum_{k=0}^{N-1} \csc\left(\frac{\pi(k + \frac{1}{2})}{N}\right).$$

Then $\|f_N\|_{W(\mathbb{T}^2)} \geq S_N$ and

$$\frac{S_N}{\|f_N\|_{H^1(\mathbb{T}^2)}} \longrightarrow \frac{\pi}{2}.$$

4 Proof

We first record the finite orthogonality identity responsible for the certificate.

Lemma 1. *For $N \geq 2$, the real $N \times N$ matrix*

$$C_{r,s} = \frac{1}{N \sin(\pi(r + s + \frac{1}{2})/N)}, \quad 0 \leq r, s \leq N-1,$$

is orthogonal.

Proof. Use the standard identities, obtained by logarithmically differentiating the finite sine product,

$$\begin{aligned}\sum_{s=0}^{N-1} \cot\left(x + \frac{s\pi}{N}\right) &= N \cot(Nx), \\ \sum_{s=0}^{N-1} \csc^2\left(x + \frac{s\pi}{N}\right) &= N^2 \csc^2(Nx).\end{aligned}$$

For distinct rows $r \neq t$, set $u_s = \pi(r + s + \frac{1}{2})/N$ and $v_s = \pi(t + s + \frac{1}{2})/N$. Since

$$\frac{1}{\sin u_s \sin v_s} = \frac{\cot u_s - \cot v_s}{\sin(v_s - u_s)},$$

the inner product of rows r and t vanishes: both cotangent sums are $N \cot(\pi(r + \frac{1}{2})) = N \cot(\pi(t + \frac{1}{2})) = 0$. For a row with itself, the second identity gives

$$\sum_{s=0}^{N-1} \csc^2 u_s = N^2 \csc^2(\pi(r + \frac{1}{2})) = N^2.$$

The prefactor N^{-2} completes the proof. □

Now put $m = N - 1$ and consider the m -homogeneous symbol

$$\psi_N(z_1, z_2) = \sum_{k=0}^m a_k z_1^k z_2^{m-k}, \quad a_k = \frac{1}{N \sin(\pi(k + \frac{1}{2})/N)}.$$

The dual of $W(\mathbb{T}^2)$ is isometric to the bounded small Hankel operators [1]. For completeness, on the homogeneous input space of degree n , $0 \leq n \leq m$, the matrix of the Hankel form associated with ψ_N is

$$A_n = (a_{i+j})_{\substack{0 \leq i \leq m-n \\ 0 \leq j \leq n}}.$$

Indeed, the product of the basis monomials $z_1^j z_2^{n-j}$ and $z_1^i z_2^{m-n-i}$ has z_1 -degree $i + j$. The matrix A_n is exactly the submatrix of C formed by rows $0, \dots, m - n$ and columns $0, \dots, n$. Lemma 1 therefore gives $\|A_n\|_{\text{op}} \leq 1$ for every n . The full Hankel operator is the orthogonal direct sum of these blocks (and is zero on input degrees larger than m), so

$$\|\psi_N\|_{W(\mathbb{T}^2)^*} \leq 1.$$

Duality now yields

$$\|f_N\|_{W(\mathbb{T}^2)} \geq |\langle f_N, \psi_N \rangle| = \sum_{k=0}^{N-1} a_k = S_N. \quad (1)$$

It remains to compare two elementary logarithmic asymptotics. Set $x_k = (k + \frac{1}{2})/N$. The function

$$h(x) = \csc(\pi x) - \frac{1}{\pi x} - \frac{1}{\pi(1-x)}$$

extends continuously to $[0, 1]$. Consequently, midpoint summation and a change of index give

$$\begin{aligned} S_N &= \frac{1}{N} \sum_{k=0}^{N-1} \left(\frac{1}{\pi x_k} + \frac{1}{\pi(1-x_k)} + h(x_k) \right) \\ &= \frac{2}{\pi} \sum_{k=0}^{N-1} \frac{1}{k + \frac{1}{2}} + O(1) = \frac{2}{\pi} \log N + O(1). \end{aligned} \quad (2)$$

On the torus,

$$|f_N(e^{i\theta_1}, e^{i\theta_2})| = \left| \sum_{k=0}^{N-1} e^{ik(\theta_1 - \theta_2)} \right|.$$

Hence normalized Haar integration and the Dirichlet-kernel formula give

$$\|f_N\|_{H^1(\mathbb{T}^2)} = \frac{1}{\pi} \int_0^\pi \frac{|\sin(Nx)|}{\sin x} dx = \frac{2}{\pi} \int_0^{\pi/2} \frac{|\sin(Nx)|}{\sin x} dx. \quad (3)$$

On $(0, \pi/2]$, the difference $1/\sin x - 1/x$ is bounded by a constant times x . Replacing $1/\sin x$ by $1/x$ in (3) therefore changes the integral by $O(1)$. After $u = Nx$,

$$\|f_N\|_{H^1(\mathbb{T}^2)} = \frac{2}{\pi} \int_0^{N\pi/2} \frac{|\sin u|}{u} du + O(1).$$

The periodic function $|\sin u| - 2/\pi$ has mean zero and a bounded primitive. Integration by parts on $[1, R]$ thus shows

$$\int_0^R \frac{|\sin u|}{u} du = \frac{2}{\pi} \log R + O(1).$$

Therefore

$$\|f_N\|_{H^1(\mathbb{T}^2)} = \frac{4}{\pi^2} \log N + O(1). \quad (4)$$

Combining (1)–(4) proves

$$\liminf_{N \rightarrow \infty} \frac{\|f_N\|_{W(\mathbb{T}^2)}}{\|f_N\|_{H^1(\mathbb{T}^2)}} \geq \frac{(2/\pi)}{(4/\pi^2)} = \frac{\pi}{2},$$

which proves Theorem 1.

5 Verification notes

The included script `code/verify_cosecant_certificate.py` checks, for several N , that CC^T is the identity to floating-point precision, every block A_n has operator norm at most one, and the numerical certificate ratio approaches $\pi/2$. These tests are sanity checks only; Lemma 1 and the asymptotic calculation are complete analytic proofs.

The most important human checks are:

1. the convention for the small Hankel pairing and the indices of A_n ;
2. that all homogeneous blocks, including the endpoint degrees, are covered by the submatrix argument;
3. the normalization of Haar measure in the Dirichlet-kernel asymptotic.

6 Limitations and novelty status

The theorem only bounds a hypothetical embedding constant. It does not show that K_2 is finite, so the source problem remains open. A digit-separated lacunary product test did not amplify the finite gain: extra Hankel interactions increased the dual norm faster than the observed H^1 gain.

A bounded search on 21 July 2026 inspected the source paper and Brevig’s earlier paper [2]. It also checked arXiv:2101.00763 and exact arXiv/web queries containing “polydisc Nehari”, “cosecant matrix”, “Dirichlet kernel”, and “ $\pi/2$ ”. No matching construction or bound was found. The $N = 3$ member of the family recovers the earlier quadratic certificate in [2], which is evidence that the present construction is a natural asymptotic extension, not evidence of definitive novelty. Novelty confidence is therefore moderate and a specialist literature review is recommended.

References

- [1] K. Bampouras and O. F. Brevig, *Norm attaining vectors and Hilbert points*, arXiv:2310.20346 (2023; revised 2024).
- [2] O. F. Brevig, *Minimal norm Hankel operators*, Proc. Amer. Math. Soc. **150** (2022), 4383–4391; arXiv:2107.01680.
- [3] J. Ortega-Cerdà and K. Seip, *A lower bound in Nehari’s theorem on the polydisc*, J. Anal. Math. **118** (2012), 339–342.